ASSIGNMENT 6 — QUESTION 4

Q4 - DevOps Delay: Async Timeout Race

JavaScript (ES6+) solution with comments

```
// Q4: Compare Promise.all() and Promise.race().
// Server A takes 2 seconds; Server B takes 3 seconds.
// Each server has a small chance of failing.

function deployServer(name, seconds) {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            if (Math.random() < 0.15) {
                reject(new Error(name + " deployment failed."));
            } else {
                resolve(name + " deployment completed");
            }
        }, seconds * 1000);
    });
}

// Promise.all resolves only when both deployments succeed.
// If either rejects, Promise.all rejects.
Promise.all([
    deployServer("Server A", 2),
    deployServer("Server B", 3)
])
    .then(() => console.log("Deployment completed for all servers"))
    .catch(error => console.error("Deployment error:", error.message));

// Promise.race settles as soon as the first Promise settles.
// The first result could be a success OR a failure.
Promise.race([
    deployServer("Server A", 2),
    deployServer("Server B", 3)
])
    .then(result => console.log("Fastest response:", result))
    .catch(error => console.error("Fastest response failed:", error.message));

/*
Note: The two examples start their own server simulations.
Therefore, their random outcomes are independent.
*/
```